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ORIGINAL ARTICLE

State-of-the-art machine learning algorithms for the prediction of outcomes after contemporary heart transplantation: Results from the UNOS database

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Abstract

Purpose: We sought to develop and validate machine learning (ML) models to increase the predictive accuracy of mortality after heart transplantation (HT).

Methods and results: We included adult HT recipients from the United Network for Organ Sharing (UNOS) database between 2010 and 2018 using solely pre-transplant variables. The study cohort comprised 18 625 patients (53 ± 13 years, 73% males) and was randomly split into a derivation and a validation cohort with a 3:1 ratio. At 1-year after HT, there were 2334 (12.5%) deaths. Out of a total of 134 pre-transplant variables, 39 were selected as highly predictive of 1-year mortality via feature selection algorithm and were used to train five ML models. AUC for the prediction of 1-year survival was .689, .642, .649, .637, .526 for the Adaboost, Logistic Regression, Decision Tree, Support Vector Machine, and K-nearest neighbor models, respectively, whereas the Index for Mortality Prediction after Cardiac Transplantation (IMPACT) score had an AUC of .569. Local interpretable model-agnostic explanations (LIME) analysis was used in the best performing model to identify the relative impact of key predictors. ML models for 3- and 5-year survival as well as acute rejection were also developed in a secondary analysis and yielded AUCs of .629, .609, and .610 using 27, 31, and 91 selected variables respectively.

Conclusion: Machine learning models showed good predictive accuracy of outcomes after heart transplantation.

KEYWORDS
graft failure, heart transplantation, machine learning, UNOS

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1 | INTRODUCTION

Heart failure is a major public health issue affecting more than 6.5 million adults in the United States.[1] In the last two decades, guideline-directed medical therapy and durable mechanical support have improved the survival of patients with heart failure.[2,3] However, heart transplantation (HT) remains the treatment of choice for patients with medically refractory end-stage heart failure given its superior long term outcomes and quality of life.[4] Today, optimization of outcomes and risk stratification are recognized as critically important issues in HT.[5]

Several risk scores have been developed for the prediction of outcomes based on traditional statistical analysis, however, their accuracy is modest.[6–8] Machine learning (ML) algorithms on the other hand are being increasingly applied in cardiology to address complex clinical issues,[9,10] and have been successfully used in predicting outcomes and discovering phenotypes in patients with heart failure.[11,12] Recently, the application of ML algorithms to predict outcomes after HT has shown some promising results compared to traditionally derived risk scores.[13,14] However, further work is needed to determine the role of ML in the prediction of outcomes and risk stratification of HT patients.

In the current study, we used data from patients undergoing HT in the United States to create and validate state-of-the-art ML algorithms for the prediction of 1-year survival after HT. A subgroup analysis on the prediction of 3- and 5-year survival as well as acute rejection was also performed.

2 | METHODS AND MATERIALS

2.1 | Data source

Data were obtained from the United Network for Organ Sharing (UNOS) database which collects all data related to waiting list registrations and transplantation that have been listed or performed in the United States and reported to the Organ Procurement and Transplantation Network (OPTN) since October 1, 1987. Extensive baseline data, including demographics, clinical and laboratory details at the time of listing and transplantation as well as follow up survival data were collected for each patient. For this analysis, we queried the UNOS database for patients that were enlisted and underwent heart transplant between 2010 and 2018.

2.2 | Study design

First, data were extracted from the UNOS database and pre-processed in order to obtain the study dataset using the higher-level programming language MATLAB. We excluded patients with multiple transplants during the index hospitalization, non-adult patients, and patients with less than 1-year follow up. We also did not analyze characteristics that had > 10% missing values. The study dataset was then randomly split into training and validation cohorts with a 3:1 ratio, respectively. These cohorts were used to train and validate ML models for 1-year survival (Figure 1). Patients with available longer follow up were used to predict 3- and 5-year survival using a similar methodology in a secondary analysis. Prediction of acute rejection within the 1st year after HT was also performed. Because the training cohort was unbalanced in regards to outcomes (i.e., more patients survived than died), a random oversampling technique was applied to obtain equal representation (i.e., alive vs. dead and acute rejection vs. no rejection), by simply replicating patients until group sizes were equal. Next, a feature selection method was used to distinguish highly informative baseline variables in regard to the outcome that were subsequently used to train five different ML algorithms. After training of the ML algorithms, the validation cohort was used to internally test the ML algorithms and determine their predictive accuracy. We separately applied the Index for Mortality Prediction after Cardiac Transplantation (IMPACT) score[6] for the prediction of 1-year survival after HT.

2.3 | Feature selection and ML models

Feature selection was applied on the training dataset to, (i) identify the most informative variables, (ii) reduce the complexity, and (iii) improve the performance of the ML models. In this study, we used the novel feature selection method SVM-FuzCoC to select an informative, non-redundant feature subset. This method is based on a combination of Support Vector Machines (SVM) with Fuzzy Logic and can provide a reasonable trade-off between accuracy and computational complexity.[15,16]

Five state-of-the-art supervised ML algorithms were evaluated for binary classification in the current study: Adaboost,[17] SVM,[18] Decision Tree (DT),[19] K-nearest neighbor (KNN),[20] and Logistic Regression. All categorical and textual variables were converted to a standardized numerical format, a common requirement for many ML algorithms. Training of the models was controlled by hyperparameter optimization, a technique used to optimize the performance of models and to avoid overfitting and bias errors. Each model was optimized with respect to a number of preselected hyperparameters.

2.4 | Accuracy and explainabilty of ML models

The predictive accuracy of the ML models was evaluated by measuring the total area under the curve (AUC) of the receiver-operator curve (ROC) as well as sensitivity, specificity, positive, and negative predictive value. ML can result in models of improved predictive accuracy that, however, do not rely in clinically meaningful predictors. We therefore performed an explainability analysis according to the method of local interpretable model-agnostic explanations (LIME) developed by Ribeiro et al.[21] to rank features in terms of their impact on 1-year survival. LIME analysis essentially approximates any complex, generalized ML prediction model with local, linear models that explain individual predictions.

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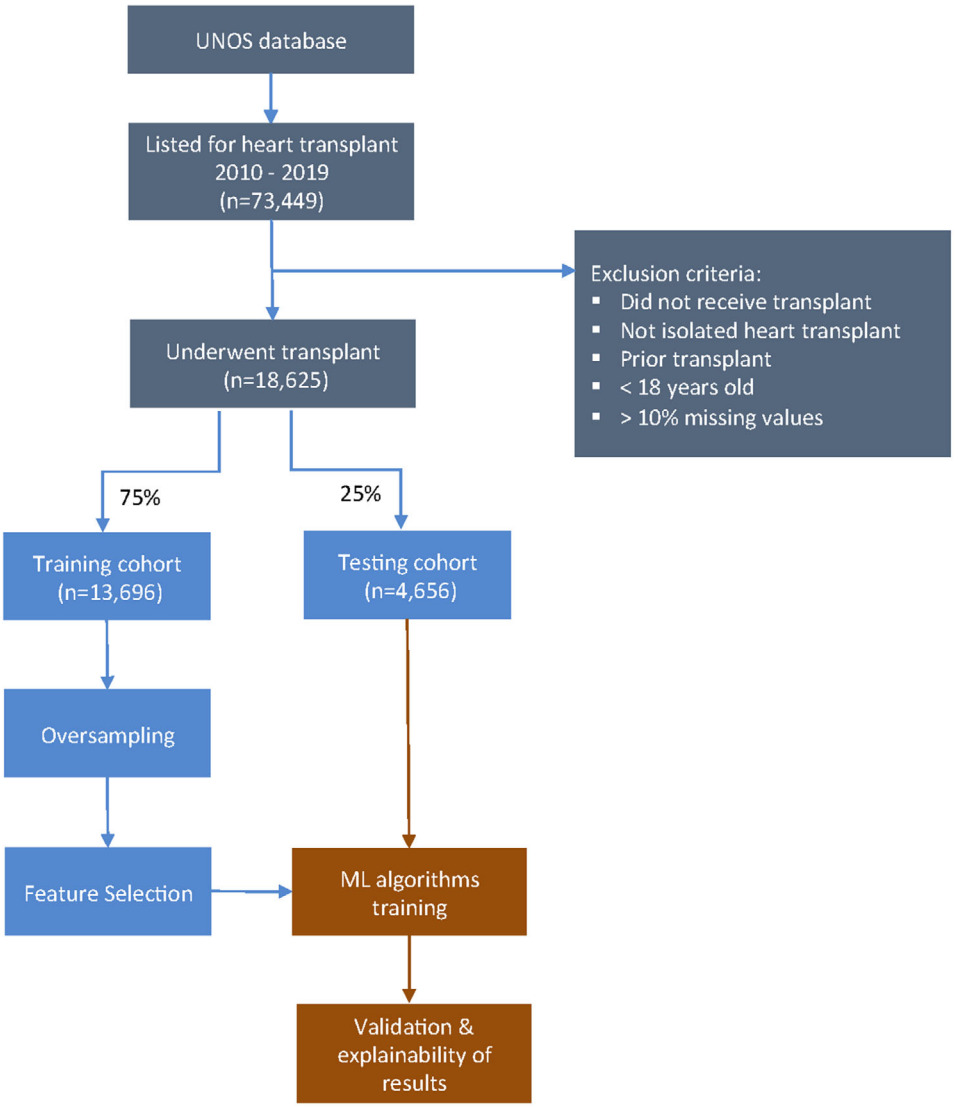

FIGURE 1 Study design for prediction of 1-year survival after heart transplantation using machine-learning algorithms. Data from the UNOS database were extracted in SAS. After applying inclusion and exclusion criteria, the dataset was randomly split in a training (75% of the dataset, $n = 13\,696$) and a validation cohort (25% of the dataset, $n = 4656$). Oversampling was performed for the patients who did not survive 1-year after transplant to allow for balanced input in the ML algorithms. Features were selected using state-of-the-art algorithms and then used to train the ML algorithms. The resulting models were tested using the validation cohort. ML, machine learning.

2.5 | Statistical analysis

Continuous and categorical variables were compared using the twotailed Student's t-test and the Chi-square exact test, respectively. Statistical analysis was performed using MATLAB 2020b (version 9.9) and STATA MP 13.0. We considered *P*-values < .05 as statistically significant.

3 | RESULTS

3.1 | Baseline characteristics

Baseline clinical and laboratory characteristics of HT recipients and their donors are shown in Table 1. The dataset of eligible patients consisted of 18 625 recipients (mean age 53 ± 13 years, 73% males) that underwent isolated HT between 2010 and 2018. At 1 year after HT, there were 2334 (12.5%) deaths. Recipients who died within the first year were older than 65 years (20% vs. 17%), slightly heavier (BMI 28 ± 5 kg/m^2 vs. 27 ± 5 kg/m^2), had a history of prior cardiac surgery (47% vs. 40%) and were more likely to have congenital heart disease (4% vs. 3%) and less likely to have dilated cardiomyopathy (80% vs. 84%) as their primary diagnosis. They were also more likely to have required mechanical ventilation (22% vs. 20%) and interestingly, less likely to have been on inotropic support (34% vs. 36%) prior to transplant. Serum creatinine prior to transplant was significantly more elevated in patients who did not survive 1-year after HT (1.4 ± .6 mg/dl vs. 1.2 ± .7 mg/ dl). They also tend to have been longer on status 1A compared (38 ± 72 days vs. 36 ± 63 days). Recipients who did not survive

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TABLE 1 Donor and recipient baseline clinical characteristics in the overall cohort as well as in the development/validation and alive/deceased at 1-year sub-cohorts

	Overall (*N* = 18 625)	Development cohort (*n* = 13 969)	Validation Cohort (*n* = 4656)	*P* value	Alive (*n* = 16 291)	Deceased (*n* = 2 334)	*P* value
Donor characteristics							
Age (years)	33 (11)	33 (11)	33 (11)	.74	32 (11)	33 (12)	<.001
Female (sex)	4981(26)	3435 (25)	1221 (26)	.95	4344	637	.52
Donor ethnicity							
White	11 976 (64)	9003 (64)	2973 (64)	.82	10 791 (66)	1539 (66)	.59
Black	3036 (16)	2262 (16)	774 (16)		3436 (21)	498 (21)	
Hispanic	2983 (16)	2210 (16)	773 (16)		1295 (8)	202 (9)	
Asian	336 (2)	258 (2)	78 (2)		565 (3)	73 (3)	
BMI (kg/m^2)	28 (6)	27 (6)	28 (6)	.30	27 (6)	28 (6)	.08
History of cocaine abuse	3604 (19)	2714 (19)	890 (19)	.59	3136 (19)	468 (20)	.47
History of alcohol abuse	3033 (16)	2305 (17)	728 (17)	.16	12 856 (79)	1829 (78)	.47
Donor cause of death							
Anoxia	4399 (24)	3300 (24)	1099 (24)	.97	3876 (24)	525 (22)	<.01
Stroke	3025 (16)	2281 (16)	744 (16)		2638 (16)	387 (17)	
Head trauma	1804 (10)	1354 (10)	450 (10)		1620 (10)	184 (8)	
Other	9387 (50)	7026 (50)	2361 (50)		8148 (50)	1239 (53)	
HTN requiring medications	5812 (31)	4332 (31)	5812 (31)	.33	5065 (31)	747 (32)	.37
On inotropic support	7906 (42)	5939 (43)	1967 (42)	.72	6934 (43)	973 (42)	.33
Serum creatinine (mg/dl)	1.5 (1.5)	1.5 (1.5)	1.4 (1.5)	.57	1.4 (1.5)	1.5 (1.5)	.32
SGPT (mg/dl)	111 (318)	112 (318)	109 (318)	.54	111 (318)	113 (319)	.81
PCO_2 (mm Hg)	38 (7)	38 (7)	38 (7)	.48	38 (7)	38 (7)	.61
Distance of donor to transplant center (miles)	169 (208)	171 (208)	164 (207)	.06	169 (208)	171 (205)	.63
Ischemic time (h)	3.1 (1)	3.1 (1)	3.1 (1)	.23	3.1 (1)	3.3 (1)	<.001
Recipient characteristics							
Age (years)	53 (13)	53 (13)	53 (13)	.32	53 (13)	54 (13)	<.001
< 65 years,	15 368 (83)	11 538 (83)	3830 (82)		13 501 (83)	1866 (80)	
≥65 years	3257 (17)	2431 (17)	826 (18)		2790 (17)	466 (20)	
Female (sex)	4981 (27)	3760 (27)	1221 (26)	.36	4344 (27)	637 (27)	.52
BMI (kg/m^2)	27.28	27.26	27.32	.47	27 (5)	28 (5)	<.001
Height (cm)	174 (10)	174 (10)	174 (10)	.28	174 (10)	173 (10)	<.001
Recipient ethnicity							
White	12 330 (66)	9234 (66)	3096 (66)	.88	10 791 (66)	1539 (66)	.79
Black	3934 (21)	2947 (21)	987 (21)		3436 (21)	498 (21)	
Hispanic	1497 (8)	1127 (8)	370 (8)		1295 (8)	202 (8)	
Asian	638 (3)	488 (3)	150 (3)		565 (3)	73 (3)	
Education							
High school	12 802 (69)	9650(69)	3152 (69)	.36	11 213 (69)	1589 (98)	.09
College	579 (3)	414 (3)	165 (3)		483 (3)	96 (4)	
Higher	27 (.1)	22 (.1)	5 (.1)		24 (.1)	3 (.1)	
Unknown/NA	5217 (28)	3883 (28)	1334 (28)		4,571 (28)	646 (28)	
Prior cardiac surgery	4875 (26)	3720 (27)	1155 (25)	.64	6444 (40)	1098 (47)	<.001

(Continues)

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TABLE 1 (Continued)

	Overall (*N* = 18 625)	Development cohort (*n* = 13 969)	Validation Cohort (*n* = 4656)	*P* value	Alive (*n* = 16 291)	Deceased (*n* = 2 334)	*P* value
Diabetes	5181	3882	1299	.48	4477 (27)	704 (30)	.12
Serum creatinine (mg/dl)	1.3 (.7)	1.3 (.7)	1.3 (.7)	.75	1.2 (.7)	1.4 (.6)	<.001
HIV +	53 (.3)	43 (.3)	10 (.3)	.31	42 (.3)	11 (.4)	.07
CMV +	10 657 (57)	7994 (57)	2663 (57)	.97	9299 (57)	1358 (58)	.31
EBV +	14 897 (80)	11 125 (80)	3772 (81)	.04	13 032 (80)	1865 (80)	.92
HBV +	7 (.1)	3 (.1)	4 (.1)	.60	5 (.1)	2 (.1)	.55
ABO match	15 978 (86)	11 970 (86)	4008 (86)	.51	14 000 (86)	1978 (85)	.12
Primary diagnosis							
Dilated cardiomyopathy	15 578 (84)	11 702 (84)	3876 (84)	.40	13 701 (84)	1877 (80)	<.001
Coronary artery disease	781 (4)	585 (4)	196 (4)	.95	675 (4)	106 (5)	.37
Restrictive cardiomyopathy	597 (3)	455 (3)	142 (3)	.49	515 (3)	82 (4)	.37
Hypertrophic cardiomyopathy	460 (2)	350 (2)	110 (2)	.59	397 (2)	63 (3)	.45
Congenital heart disease	527 (3)	388 (3)	139 (3)	.46	425 (3)	102 (4)	<.001
Valvular cardiomyopathy	245 (1)	194 (1)	51 (1)	.11	229 (1)	23 (1)	.10
On inotropes	6712 (36)	5000 (36)	1712 (36)	.23	5923 (36)	789 (34)	.02
On prostaglandins	67 (.4)	59 (.4)	8 (.4)	.66	56 (.3)	11 (.5)	.39
Mechanical ventilation	3718 (20)	2809 (20)	909 (20)	.39	3203 (20)	515 (22)	<.01
Ventricular assist device							
None	13 170 (71)	9876 (71)	3294 (71)	.57	11 582 (71)	1588 (68)	<.001
LVAD	5071 (27)	3795 (27)	1276 (27)		4402 (27)	669 (29)	
RVAD	12 (.1)	11 (.1)	1 (.1)		9 (.1)	3 (.1)	
TAH	104 (.6)	82 (.6)	22 (.6)		73 (.4)	31 (.1)	
LVAD/RVAD	267 (1)	204 (1)	63 (1)		224 (1)	42 (2)	
ICD	14 340 (77)	10 718 (77)	3622 (77)	.14	2455 (15)	373 (16)	.08
Year of transplant							
2010	1767 (9)	1332 (9)	435 (9)	.89	1607 (10)	160 (7)	<.001
2011	1747 (9)	1302 (9)	445 (9)		1569 (9)	178 (8)	
2012	1786 (10)	1354 (10)	432 (10)		1603 (10)	183 (8)	
2013	1893 (10)	1411 (10)	482 (10)		1711 (10)	182 (8)	
2014	1993 (11)	1470 (11)	523 (11)		1782 (11)	211 (9)	
2015	2062 (11)	1546 (11)	516 (11)		1878 (12)	184 (8)	
2016	2415 (13)	1816 (13)	599 (13)		2197 (13)	218 (9)	
2017	2440 (13)	1847 (13)	593 (13)		2200 (13)	240 (10)	
2018	2522 (13)	1891 (13)	631 (13)		1744 (11)	778 (33)	
Days listed	237 (371)	237 (369)	239 (376)	.70	236 (371)	245 (370)	.30
Days as status 1A	36 (64)	36 (64)	36 (65)	.78	36 (63)	38 (72)	.06
Previous transplant	519 (3)	380 (3)	139 (3)	.37	414 (3)	105 (4)	<.001

Abbreviations: BM, body mass index; ICD; implantable cardiac defibrillator; LVAD/RVAD, left/right ventricular assist device; TAH, total artificial heart.

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TABLE 2 Baseline donor, recipient, and transplant characteristics selected by the feature selection algorithm as highly relevant to 1-year survival

	Characteristics		
	Demographic	**History/clinical**	**Laboratory**
Recipient	Age	Ventricular assist device	Creatinine (mg/dl)
	Sex	Prior cardiac surgery	HIV +
	Ethnicity	Heart failure etiology	EBV +
	Height	Waiting time in 1A status	CMV +
	BMI	Total waiting time	HBV (HBsAg+)
	Level of education	Treatment with prostaglandins	
		ICD	
		Inotropic support	
		Mechanical ventilation	
Donor	Age	Alcohol abuse	Creatinine (mg/dl)
	Ethnicity	Cocaine abuse	SGPT (mg/dl)
	BMI	Inotropic support	PCO2 (mm Hg)
		HTN requiring medical therapy	
		Cause of death	
	Characteristics		
Transplant		Ischemic time	
		Transplant surgery duration	
		Transplant year	
		Listing year	
		ABO match	

These variables were used to train the ML algorithms . Within 2010–2018.
Abbreviations: BMI, body mass index; ICD; implantable cardiac defibrillator; ML, machine learning; PCO_2, arterial partial CO_2 pressure; SGPT, alanine aminotransferase.

after 1 year were more likely to have received the graft from a slightly older donor (33 ± 12 vs. 32 ± 11 years old).

A total of 13 969 (75%) and 4656 (25%) patients were randomly assigned to the development and validation cohort, respectively. There were no significant differences between patients assigned to the development and validation cohort, with the exception of slightly more frequently EBV seropositive patients in the validation cohort (80% vs. 81%).

3.2 | ML model performance and explainability

One hundred and thirty-four pre-transplant variables were analyzed. Out of them, 39 were selected by the feature selection method as highly predictive of 1-year mortality (Table 2). The selected features were recipient and donor variables, as well as variables pertinent to transplant surgery. The predictive performance of all ML models for 1-year mortality is summarized in Figure 2. Although the best performance was achieved by Adaboost (AUC .689, 95% CI .665–.715), all ML models achieved higher AUC compared to IMPACT score with the exception of the K-nearest neighbor (SVM AUC .637 95% CI .612–.662, DT AUC .649 95% CI .622–.676, LR AUC .642 95% CI .614–.667 and KNN AUC .526 95% CI .502–.550 in contrast to IMPACT AUC .569, 95% CI .545–.592). Figure 2(B) shows the discriminatory performance of the ML models in relation to number of variables used. With the exception of K-nearest neighbor, there was an exponential increase in AUC with the use of 1–5 pre-transplant variables. However, the increase in AUC gradually plateaued after the use of 10 variables. Sensitivity, specificity, positive, and negative predictive value of the ML models using the validation sub-cohort ranged from 55.6% to 68.5%, 42.6% to 63.4%, 88.7% to 93.6% and 13.5% to 21.6%, respectively, as shown in Table 3.

The results of explainability analysis for the best ML model using LIME are shown in Figure 3. Serum creatinine of the recipient had the strongest relative impact on 1-year mortality after HT, followed by height of the recipient and ischemic time. The other top variables that were used by the ML model were recipient BMI, year of transplant, type of ventricular assist device (VAD), recipient history of prior cardiac surgery, etiology of heart failure, and recipient/donor age.

Results of the secondary analysis for prediction of 3- and 5-year survival are summarized in Table 4 and Supplemental Figure 1. AUC of the best ML model decreased to .605 and .628, respectively. Additionally, 91 variables (features) had to be incorporated in the ML model for 5-year mortality to achieve this performance. For the prediction of

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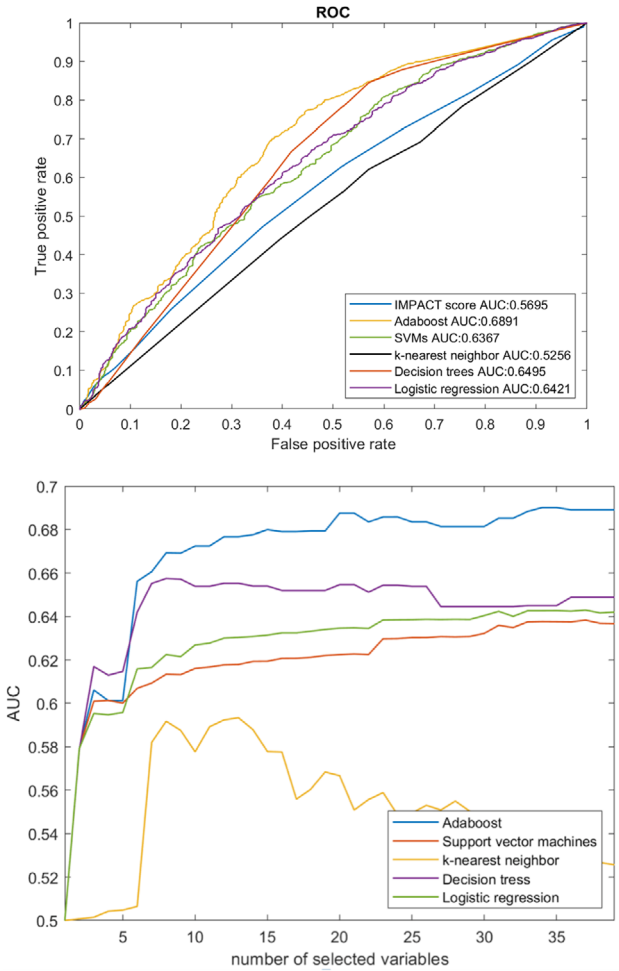

FIGURE 2 Predictive performance of ML models for 1-year mortality after HT. (A). Receiver-operating characteristic (ROC) curves of ML models using 39 selected variables and the IMPACT score. (B). Predictive performance of ML models by number of variables included. AUC, area under the curve; HT, heart transplantation; ML, machine learning; ROC, receiver-operating characteristic; SVM, support vector machine.

TABLE 3 Predictive performance of ML models in the testing (validation) cohort for 1-year mortality

ML model	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)
Adaboost	63.0	68.5	21.6	93.6
SVM	63.4	55.6	16.5	91.7
K-nearest neighbor	42.6	62.4	13.5	88.7
Decision trees	57.5	67.9	19.9	92.1
Logistic regression	61.1	59.2	17.1	91.7

Sensitivity refers to the proportion of patients with event (deceased) that were correctly identified by the ML algorithm. Specificity refers to the proportion of patients without event (alive) that were correctly identified by the ML algorithm. PPV refers to the proportion of patients that were identified by the ML algorithms as having an event, that really had an event. PPV refers to the proportion of patients that were identified by the ML algorithms as not having an event, that really did not have an event.
Abbreviations: ML, machine learning; SVM, support vector machine.

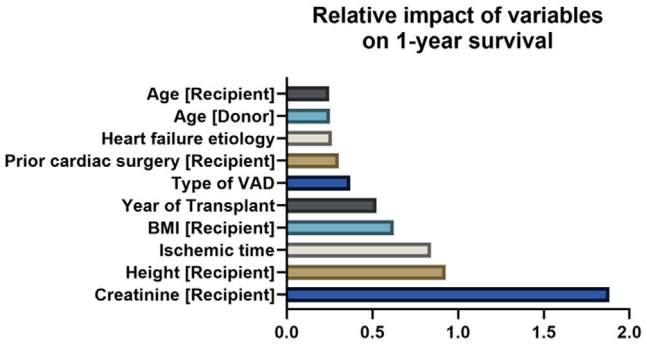

FIGURE 3 LIME analysis visualizing relative impact of top-10 variables on 1-year mortality for the best performing ML model (Adaboost). Results in logarithmic scale. LIME, local interpretable model-agnostic explanations; ML, machine learning.

TABLE 4 Predictive performance of the best ML model for mortality after HT (1, 3, and 5 years)

Prediction	Selected features	Sensitivity	Specificity	AUC
1-year survival	39	.6300	.6851	.689
3-year survival	31	.6207	.5473	.60528
5-year survival	91	.6144	.5956	.6283

Abbreviations: AUC, area under the curve; ML, machine learning.

1-year acute rejection, Adaboost again achieved the highest predictive performance (AUC .629 95% CI .617–.638) (Supplemental Figure 2) with a sensitivity and specificity of .541 and .651 respectively.

4 | DISCUSSION

The salient findings of this analysis of contemporary HT recipients from a nationwide registry with utilization of ML algorithms can be summarized as follows: (i) state-of-the-art ML models introduced in this analysis can be utilized to predict mortality after HT using solely baseline characteristics with improved accuracy compared to previously published risk scores, (ii) several established and novel donor, recipient and transplant related predictors of mortality were identified by an efficient feature selection algorithm, and (iii) LIME explainability analysis of the best ML model revealed a clinically reliable ML model and highlighted the relative impact of the strongest predictors, particularly recipient creatinine for the prediction of 1-year survival.

About 3000 HT are performed annually in the United States with 1-year survival close to 90%,[22] whereas patients who survive 1 year have a median survival of 14 years.[23] Large national and international HT registries have been used to create risk scores for HT patients. The IMPACT risk score, for example, was created using the UNOS database and was based on recipient characteristics.[6] Other risk scores have incorporated donor characteristics.[7,8] However, risk scores based on traditional statistical methods, that is, regression, achieve modest predictive accuracy at best. Artificial intelligence and ML algorithms on

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the other hand offer some striking advantages compared to traditional statistical methods in regards to their ability to analyze large and multivariable datasets with the purpose of reducing complexity and discovering potentially useful hidden associations within the data.[24] Overall, ML algorithms can be trained either unsupervised, where the ML algorithm is left to discover associations between data, or with supervision, where the training occurs based on predefined output. In the current study, we used supervised ML algorithms to predict mortality after HT using pre-transplant recipient and donor characteristics. Almost 130 variables were analyzed, a task that is challenging with traditional statistical methodology. With the exception of K-nearest neighbor, all ML algorithms in the current study showed improved predictive accuracy. Additionally, it is worth commenting on the high negative predictive value that was consistent among all ML models.

ML models have been applied in HT with promising results.[13,14] The previously published International Heart Transplant Survival Algorithm (IHTSA) was based on a flexible, nonlinear artificial neural network to predict survival after HT, included over 56 000 adult patients who underwent HT from 1994 to 2010 with a reported AUC .65 for 1-year survival.[25] The analysis identified recipient age, type of heart disease, history of prior transplantation, and mechanical ventilation before transplantation and donor age, cause of death, sex, and diabetes as important predictors of mortality. In contrast to that, we analyzed HT recipients after 2010, thus, representing contemporary practices and outcomes. Furthermore, our analysis employed efficient feature selection as a separate process prior to ML according to an established methodology,[9] which resulted in identifying predictors such as recipient age, BMI, prior cardiac surgery, etiology of heart failure, mechanical ventilation, and ischemic time. These predictors overlap with previously published analyses, including a recent in-depth survival analysis by Hsich et al.[26] However, our analysis employed an entirely ML-based methodology, in contrast to the study by Hsich et al., where ML was mostly used to analyze relations between predictors of mortality. In addition to that, our results draw attention to the potential importance of donor parameters such as history of alcohol and cocaine use, cause of death, and renal and liver indices. Given the nature of feature selection, we consider these results as hypothesis generating.

Taking a step further, we performed explainability analysis to verify that our ML models can be interpreted clinically and assess the impact of selected features on 1-year. Recipient pre-transplant serum creatinine stands out as a marker of end-organ function that is strongly associated with mortality after HT in this analysis. Many patients with baseline renal dysfunction recover their native renal function after HT with restoration of renal perfusion. However, renal dysfunction is frequent after HT and associated with baseline renal dysfunction.[27] In particular, patients with intrinsic kidney disease may experience progression of renal impairment after HT, partially due to procedure related injuries: reduced renal blood flow and calcineurin inhibitor toxicity can worsen renal function in patients with baseline abnormal renal function. Patients with renal dysfunction at the time of HT may experience challenges with initiation and maintenance of calcineurin inhibitor-based immunosuppression regimens resulting in worse renal and overall outcomes in the long-term. Review of the rest top mortality predictors verifies that our model is clinically interpretable and therefore reliable.

Results of our secondary analyses regarding prediction of 3- and 5-year mortality shows decrease in accuracy, as well as requirement of a significant number of predictors in the case of 5-year mortality. This likely reflects the inability to predict long term outcomes using pre-transplant variables only, since for example post-transplant immunosuppressive regimens were not included in our analysis. For the same reason, moderate predictive accuracy was also achieved for the prediction of acute rejection. Given the suboptimal accuracy of these ML models, these analyses are mostly exploratory and LIME analysis was not performed.

4.1 | Study limitations

The inability of the ML models to achieve far superior predictive accuracy could be attributed to data quality of the underlying dataset, in accordance to prior similar reports.[28] However, the UNOS database is a clinical database currently used broadly for epidemiologic analysis and outcomes. A second limitation is that our ML models were developed using a contemporary cohort of patients in contrast to the IMPACT score which was developed using earlier cohorts. Thus, a comparison between our results and IMPACT may not be straightforward. Our results additionally require external validation. Furthermore, we only used pre-transplant characteristics in this analysis. As a result, post-transplantation events such as renal failure, malignancy, and coronary artery vasculopathy were not included. Finally, although one of the main disadvantages of ML is the lack of transparency and explainability, this was addressed in the current study with the use of LIME analysis.

In conclusion, ML models created and validated using a contemporary cohort of the UNOS database showed improved accuracy in predicting 1-year survival after HT. Further validation of these models in other HT cohorts and the assessment of their predictive capacity of long-term survival after HT are warranted.

4.2 | Clinical use and future directions

As a next step, the authors plan to develop and launch a web tool that will allow physicians to calculate 1-year mortality for a specific heart transplant candidate based on the Adaboost model.

CONFLICT OF INTEREST

None

DATA AVAILABILITY STATEMENT

Data will be available upon request

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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